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SPATIAL ACCESSIBILITY AND GIS

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must
Enabling multispecies transitions

SUSTRA
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WHAT IS SPATIAL ACCESSIBILITY?

TAKE A MOMENT TO THINK ABOUT

How did you get here this morning?

How long did it take you?

Could you realistically take other routes or modes of transport?

Are there other places close-by where you plan to go afterwards?

DEFINING ACCESSIBILITY (IN A GEOGRAPHICAL CONTEXT)

“The potential of opportunities for interaction” (Hansen 1959)

”The degree to which two places (or points) on the same surface are connected” (Ingram 1971)

”The ease with which any land-use activity can be reached from a location using a particular transport system” (Dalvi and Martin 1976)

“The extent to which land-use and transport systems enable (groups of) individuals to reach activities or destinations by means of a (combination of) transport mode(s)”(Geurs & van Wee 2004)

“The amount and the diversity of places of activity that can be reached within a given travel time and/or cost” (Bertolini et al. 2005)

”The ability of people to reach and participate in activities” (Benenson et al. 2009)

ANALYSING ACCESSIBILITY IS INTERESTING IN MANY WAYS…

“Accessibility has become a central concept in physical planning in the past decades” (Pivoni et al. 2018)

https://royalsocietypublishing.org/doi/10.1098/rsos.171668

- Accessibility is a driving force in housing and business location decisions
- Accessibility is an indicator of land-use and transport integration
- Accessibility is a social indicator
- Accessibility is an environmental indicator

15-MINUTE CITIES

- Idea, which orinated from Paris by Carlos Moreno, and popularized during the COVID-19
- The core tenet is to ensure access to essential services to everyone within 15min walk (or bike ride)
- Potentially transformative idea has attracted planners, policy-makers and researchers
- Has brought accessibility more to the public debate as everyone can understand the concept



Source: The Urbanist
https://www.theurbanist.org/2021/08/16/study-ranks-u-s-cities-by-15-minute-city-potential/

APPROACHES TO MEASURE ACCESSIBILITY

ACCESSIBILITY COMPONENTS



1) Land use

- Amount, quality and spatial distribution of opportunities at destinations (e.g. jobs, services, leisure activities)
- Demand for opportunities
- Conflict between supply and demand



2) Transport

- Location and characteristics of the transport system between origins and destinations (e.g. roads, PT connections, bike infra)



3) Time

- Availability of opportunities at different times of the day or seasons (e.g. opening hours, PT schedules)
- Time budgets of individuals



4) People

- Needs, abilities and opportunities of individuals (e.g. vehicle ownership, income, gender, age)
- Individual preferences

Geurs & van Wee (2004). Accessibility evaluation of land-use and transport strategies: review and research directions
https://www.sciencedirect.com/science/article/pii/S0966692303000607

ACCESSIBILITY COMPONENTS



1) Land use

- Amount, quality and spatial distribution of opportunities at destinations (e.g. jobs, services, leisure activities)
- Demand for opportunities
- Conflict between supply and demand

Origins and destinations



2) Transport

- Location and characteristics of the transport system between origins and destinations (e.g. roads, PT connections, bike infra)

Transport system



3) Time

- Availability of opportunities at different times of the day (e.g. opening hours, PT schedules) or seasons
- Time budgets of individuals

Temporal constraints



4) People

- Needs, abilities and opportunities of individuals (e.g. vehicle ownership, income, gender, age)
- Individual preferences

Individual (group) characteristics

Geurs & van Wee (2004). Accessibility evaluation of land-use and transport strategies: review and research directions
https://www.sciencedirect.com/science/article/pii/S0966692303000607

MEASURES OF TRAVEL COST

Physical distance

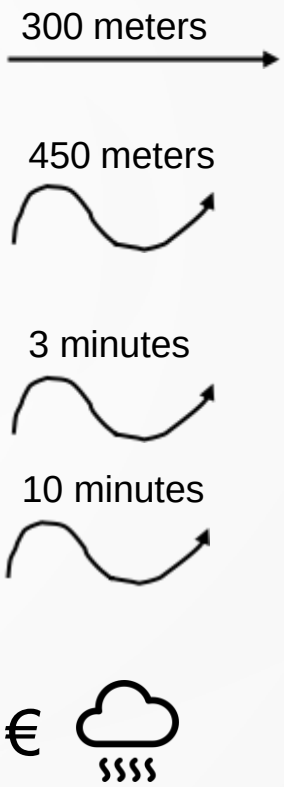
- Euclidean distance / straight-line distance / ”as-the-crow-flies”
- Network distance

Travel time

- Calculated travel time
- Perceived and experienced travel time

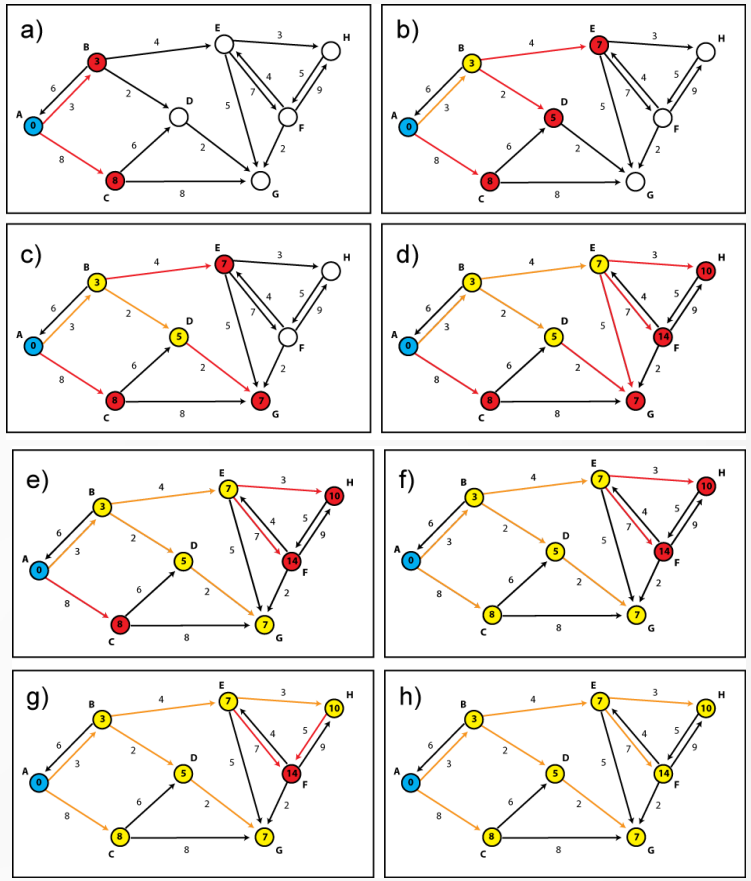
Other

- E.g. financial costs, safety/exposure costs, emissions costs



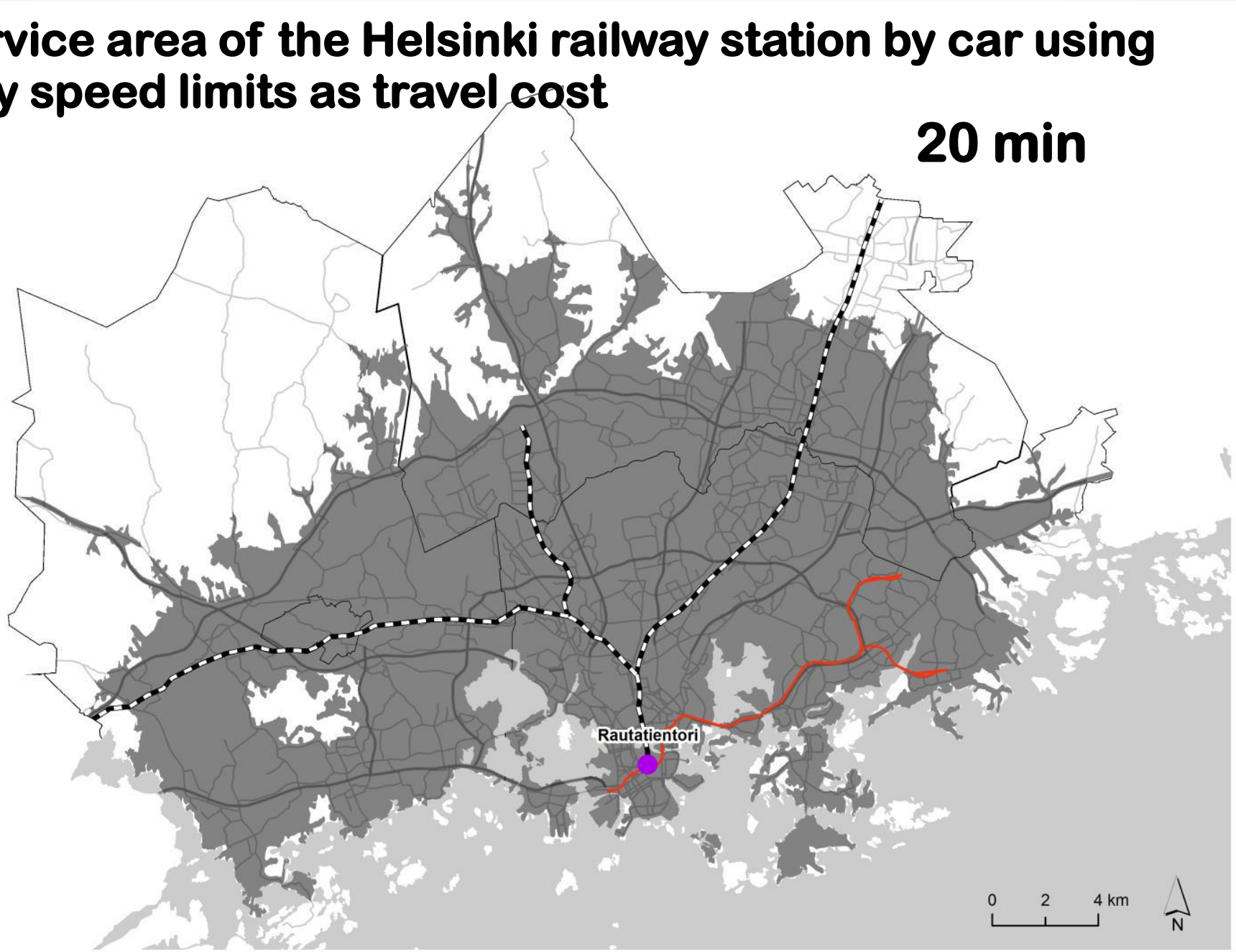
THE MOST TYPICAL APPROACH: NETWORK ANALYSIS BY PRIVATE CAR BASED ON TRAVEL TIME

- Topologically correct network dataset, consisting of Edges and Nodes
- Geometry (lengths of the road segments)
- Embedded attributes and rules
  - Turning
  - One-way streets
  - Speed-limits
- Many GIS tools and packages available for QGIS, ArcGIS, Python and R



Accessibility Component: Transport

Service area of the Helsinki railway station by car using only speed limits as travel cost



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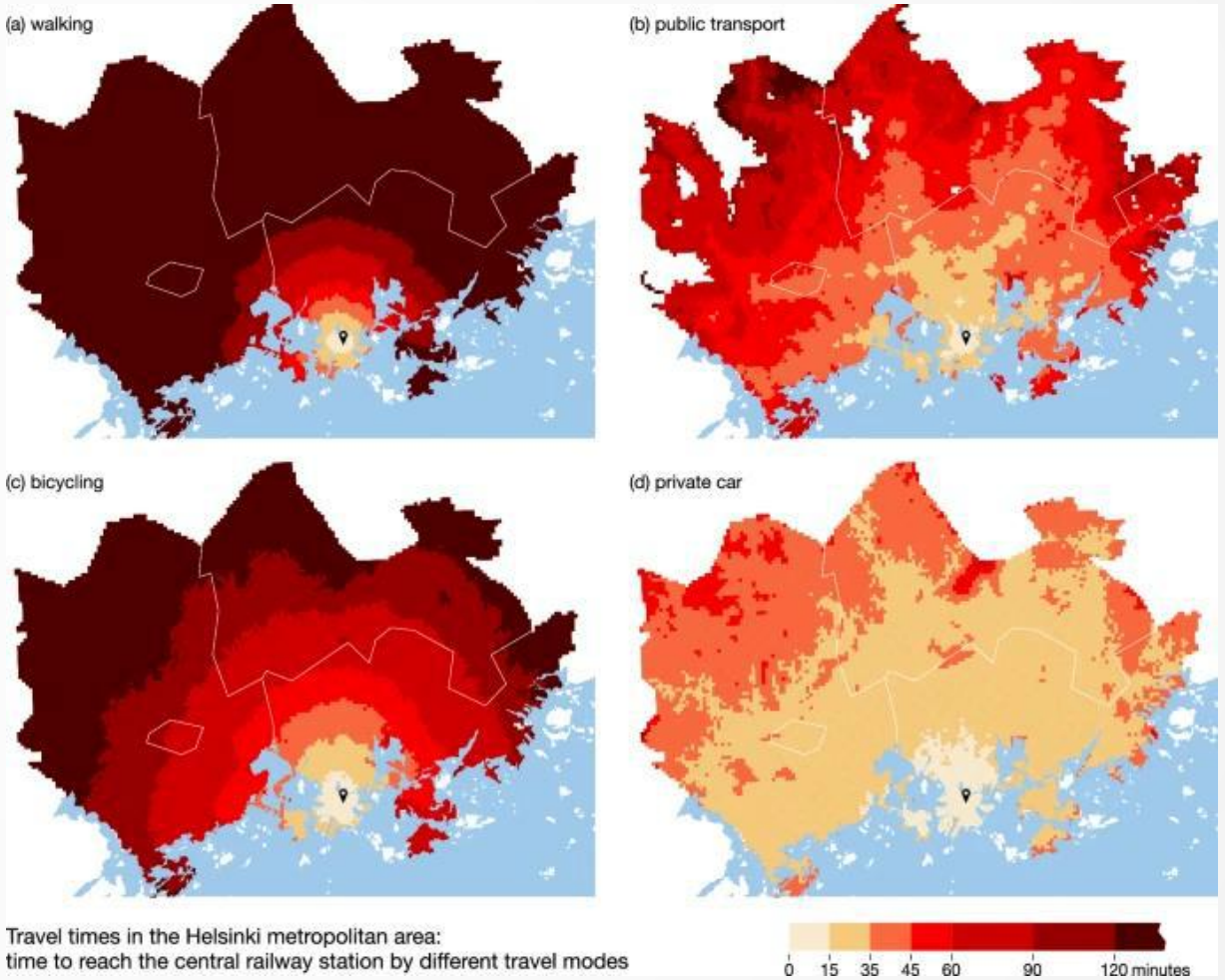
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HELSINKI REGION TRAVEL TIME MATRIX

- **Travel times** and **travel distances** in the Helsinki metropolitan area
- Published in 2013, 2015, 2018 & 2023
- 250x250m resolution, 13 132 grid cells compatible with Statistics Finland and the YKR (Yhdyskuntarakenteenseurantajärjestelmä) datasets
- Most recent iteration calculated with R5py

Fink., et al.. (2022): R5py: Rapid Realistic Routing with R5 in Python.
DOI:10.5281/ZENODO.7060437

- Widely applied in the planning organisations of the Helsinki region



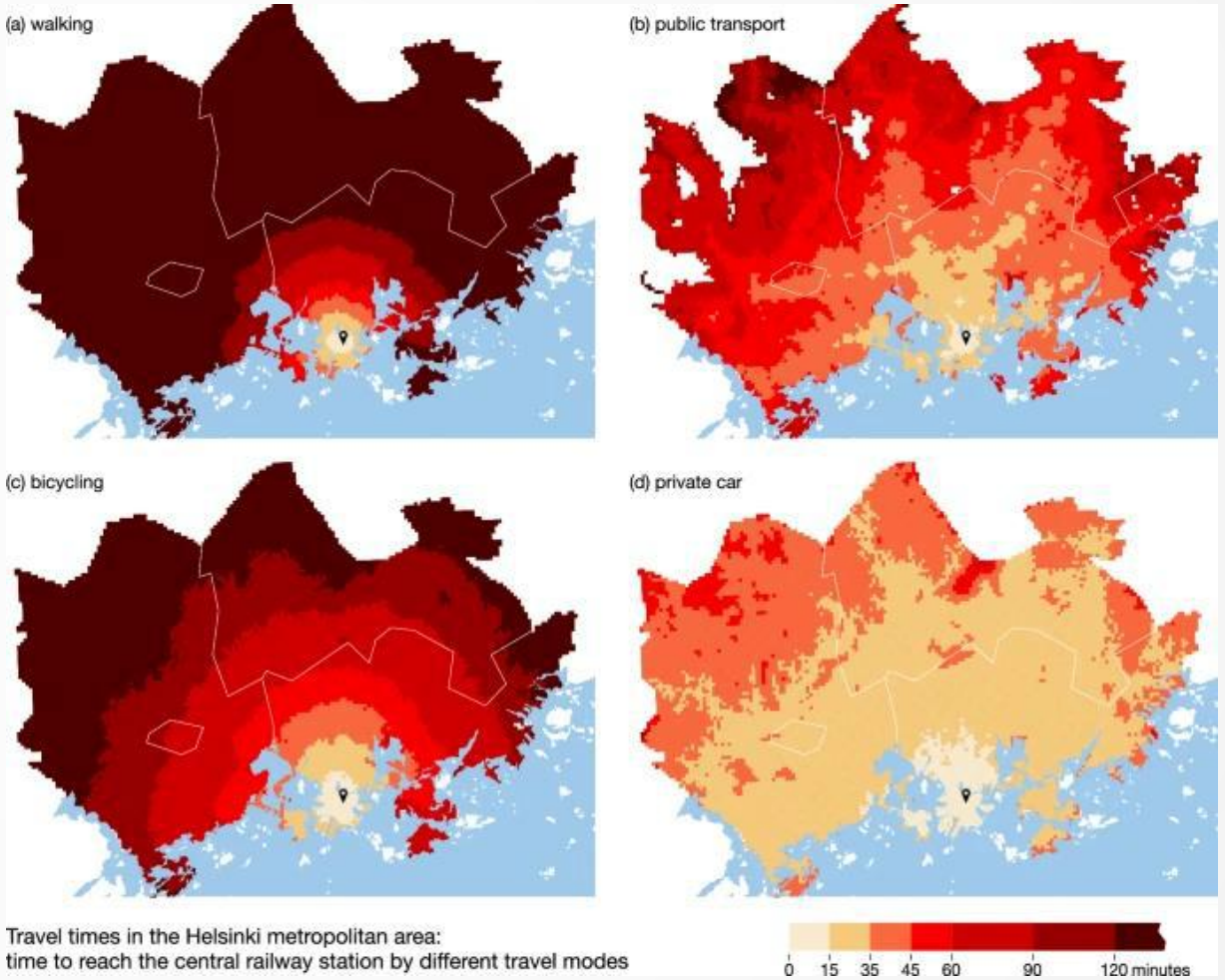
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Fink et al. (2024). A travel time matrix data set for the Helsinki region 2023 that is sensitive to time, mode and interpersonal differences, and uses open data and novel open-source software.
https://doi.org/10.1038/s41597-024-03689-z

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HELSINKI REGION TRAVEL TIME MATRIX

- **Travel modes:** car, public transport, bike, walking
- **Temporal dynamics:** includes midday, rush hour and nighttime traffic
- **Person-sensitive:** different speeds for walking and cycling



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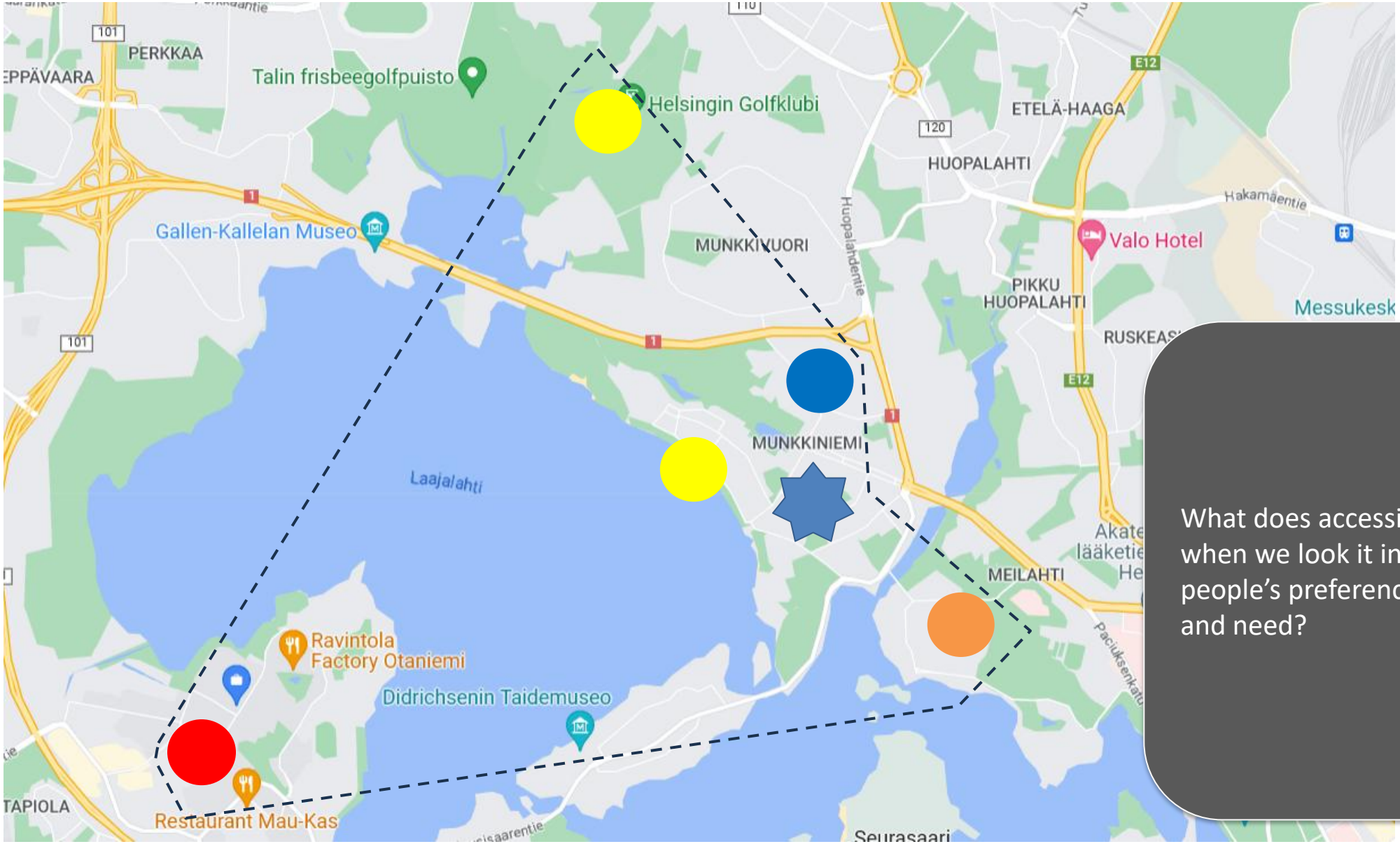
Column name	Description (times in minutes, distance is meters)
from_id:	ID number of the origin grid cell
to_id:	ID number of the destination grid cell
walk_avg:	Travel time in minutes from origin to destination by walking average speed
walk_slo	Travel time in minutes from origin to destination by walking slow speed
walk_d	Distance in meters of the walking route
bike_fst	Total travel time in minutes from origin to destination by fast cycling speed; Includes extra time (1 min) that it takes to take/return bike
bike_avg	Total travel time in minutes from origin to destination by average cycling speed; Includes extra time (1 min) that it takes to take/return bike
bike_slo	Total travel time in minutes from origin to destination by slow cycling speed; Includes extra time (1 min) that it takes to take/return bike
car_r	Travel time in minutes from origin to destination by private car in rush hour traffic; the whole travel chain has been taken into account
car_m	Travel time in minutes from origin to destination by private car in midday traffic; the whole travel chain has been taken into account
car_n	Travel time in minutes from origin to destination by private car in nighttime traffic; the whole travel chain has been taken into account
pt_r_avg:	Travel time in minutes from origin to destination by public transport in rush hour traffic by average walking speed; whole travel chain has been taken into account excluding the waiting time at home
pt_r_slo:	Travel time in minutes from origin to destination by public transport in rush hour traffic by slow walking speed; whole travel chain has been taken into account excluding the waiting time at home
pt_m_avg:	Travel time in minutes from origin to destination by public transport in midday hour traffic by average walking speed; whole travel chain has been taken into account excluding the waiting time at home
pt_m_slo:	Travel time in minutes from origin to destination by public transport in midday hour traffic by slow walking speed; whole travel chain has been taken into account excluding the waiting time at home
pt_n_avg:	Travel time in minutes from origin to destination by public transport in nighttime traffic by average walking speed; whole travel chain has been taken into account excluding the waiting time at home
pt_n_slo:	Travel time in minutes from origin to destination by public transport in nighttime traffic by slow walking speed; whole travel chain has been taken into account excluding the waiting time at home

Accessibility Component: People – Individual preferences

Kamyar Hasanzadeh

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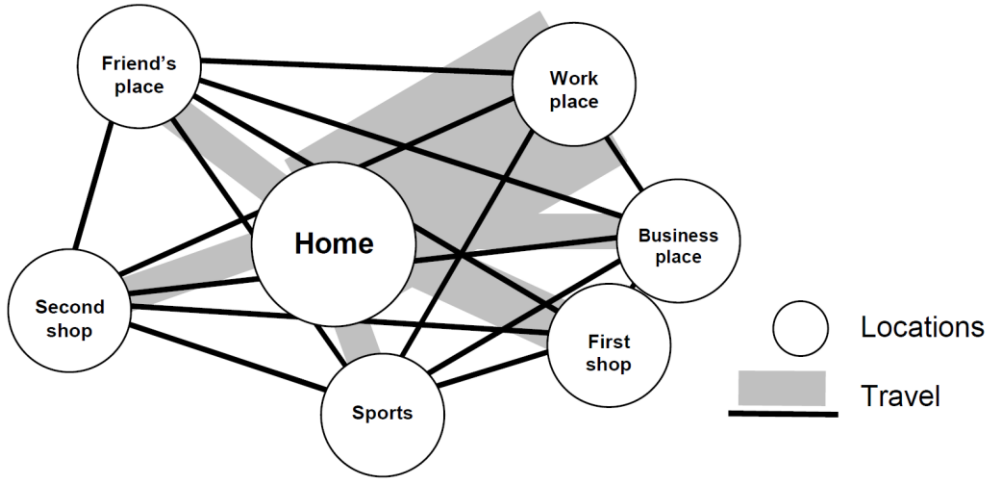
Person and place based analysis

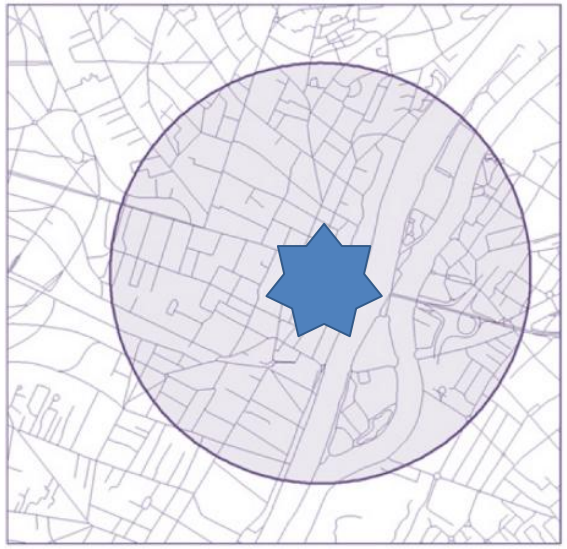


What does accessibility mean when we look it into from people's preferences, abilities and need?

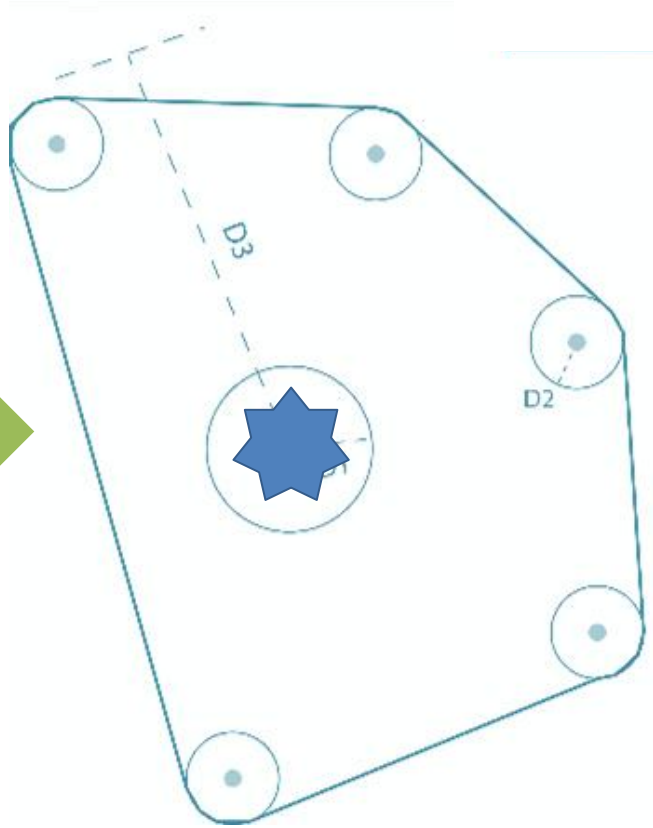
What is Activity space (AS)?

- Activity space is a set of geographically distributed locations which are physically contacted by individuals (Reynolds, 1971)
- First introduced in zoology (Burt, 1943)



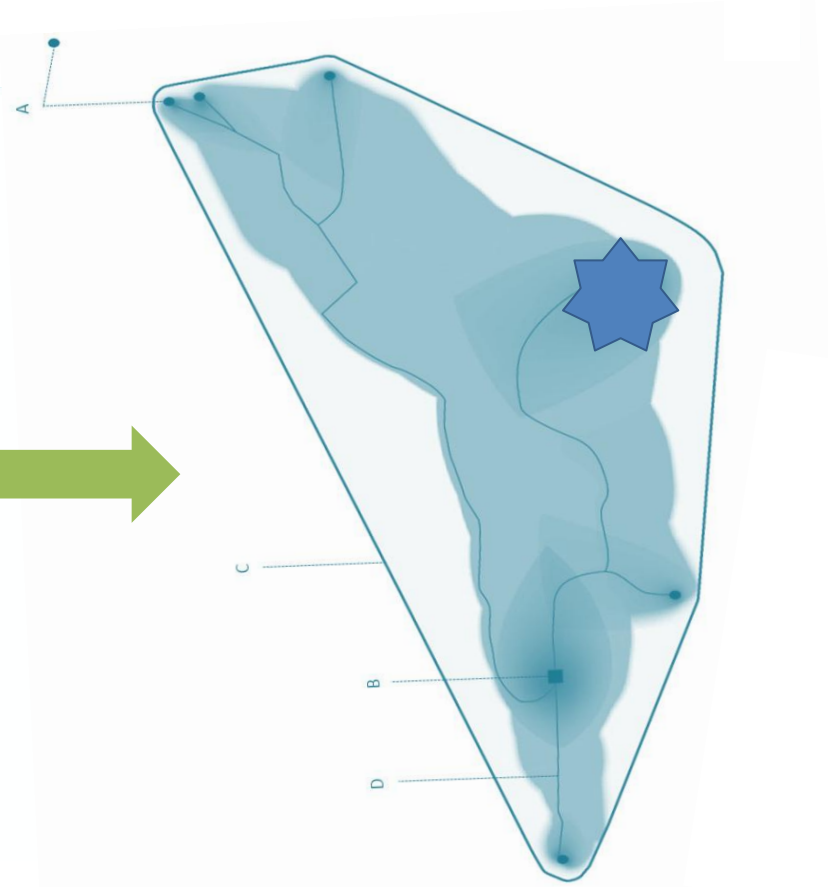


1. Static buffer around home



2. Dynamic home range model

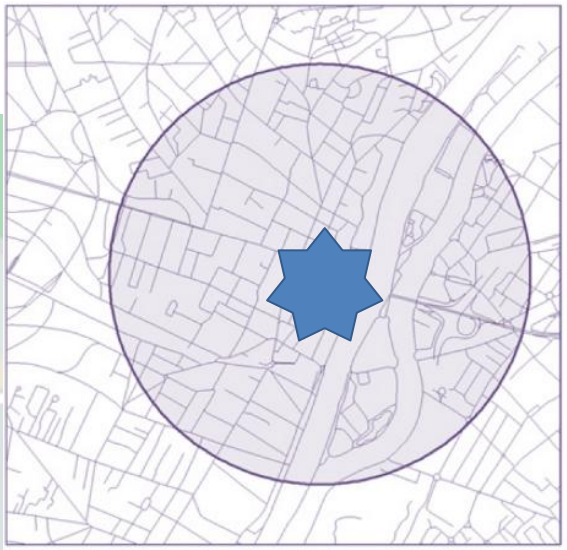
(Hasanzadeh et al., 2017)



3. Individualized residential exposure model (IREM)

(Hasanzadeh et al., 2018)

Static buffer around home, usually 500 m



Buffer fails to grasp the reality of indidual's activity space, as many errand points lie outside the buffer.

Availability does not mean accessibility.

SUSTRA
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Urban Green Accessibility

- Digital map-based surveys (Maptionaire) to map the nature places that are part of th resident’s everyday life and associated values with them.

- Survey targeted residents from Turku, Tampere and Lappeenranta.

- Respodents where asked to mark on the map
  - Everyday nature places
  - Home locations
  - Workplace, hobbies and other everyday locations

Public participation GIS (PPGIS) data



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Tampere
260 000 residents
1299 map respondents
(13% response rate)
2644 everyday nature places marked



Turku
206 000 residents
989 map respondents
(14% response rate)
1914 everyday nature places marked



Lappenranta
73 000 residents
350 map respondents
(13% response rate)
551 everyday nature places marked

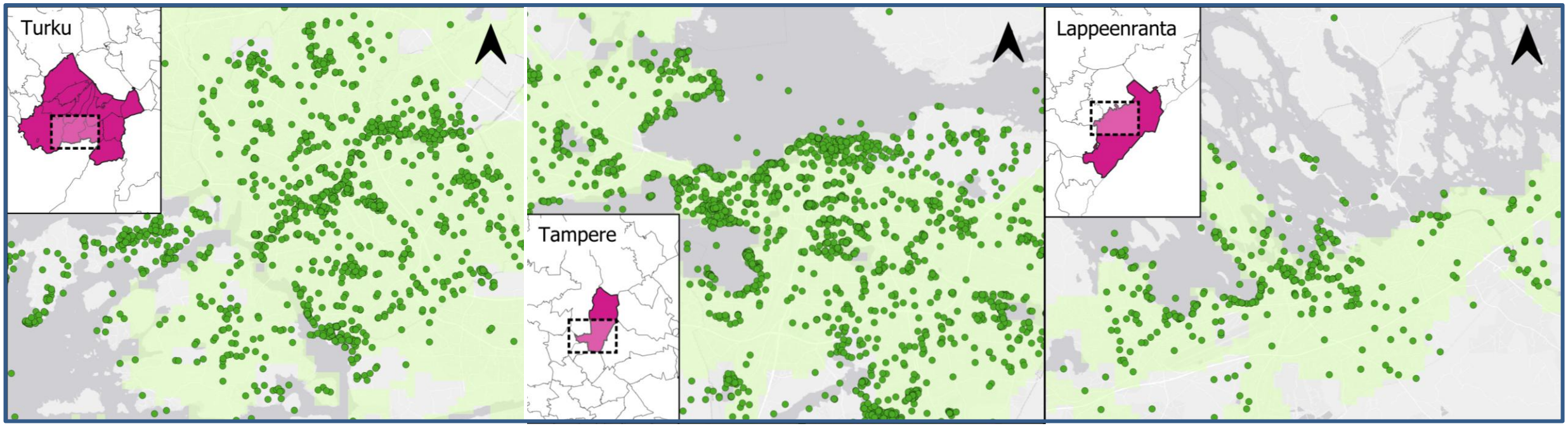
Image SourceS:
Tampere - Unknown Author is licensed under CC BY-SA

Turku - Unknown Author is licensed under CC BY-SA-NC

Lappeenranta - Unknown Author is licensed under CC BY-NC-ND

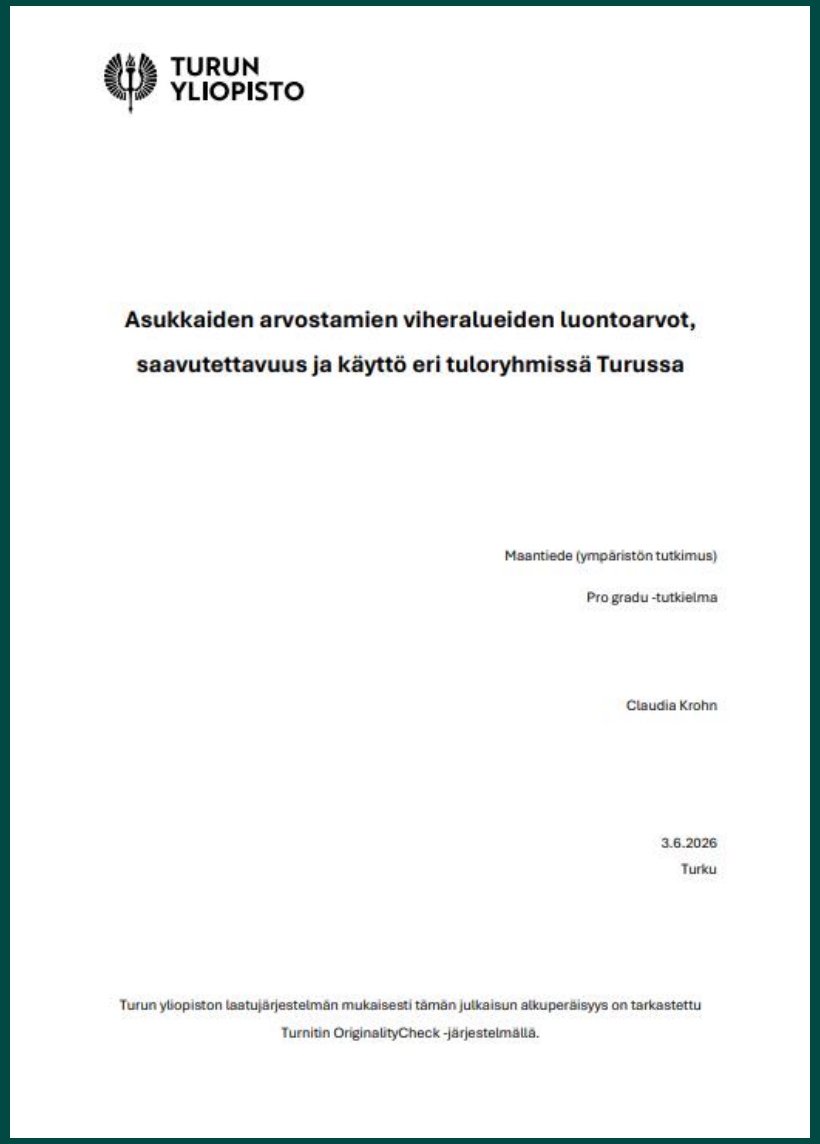
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Mapped Everday Nature Places



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TURUN YLIOPISTO

Asukkaiden arvostamien viheralueiden luontoarvot, saavutettavuus ja käyttö eri tuloryhmissä Turussa

Maantiede (ympäristön tutkimus)

Pro gradu -tutkielma

Claudia Krohn

3.6.2026
Turku

Turun yliopiston laatujärjestelmän mukaisesti tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck -järjestelmällä.

Values of urban green by residents, their use and accessibility by different income groups: Turku

Claudia Krohn, 2026

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Accessibility vs Resident's Income group

The relationship of low-income people with nature is built on local nature.

Important nature places are **closer to home** (Md = 1.33 km).

They mainly travelled to the place on **foot** (as the primary mode of transport for 58%).

Typically spend **30-60 minutes** at a time.

Commuting and passing by during everyday life is emphasized, which indicated people with low income do not always go to natural places separately, but encounter them as part of everyday mobility, more than other income groups.

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For high income earners, important nature places are located further away.

These areas are located **far away (Md = 4.27 km)** from home.

They are travelled more **often by car.**

Spend longer hours **1-2 hours.**

Engage in more **relaxing activities**, such as nature observation or gardening.

Middle-income residents consistance place between low and high income earners in all measures.

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PPGIS – Me an Nature in My Everyday Life

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PPGIS Metadata – Me an Nature in My Everyday Life

Important_nature_geopoints	Coordinate system: WGS1984	
Headline of colum	**Question**	**Answer options**
Respondent ID	Respondent ID	unique numeric identifier
Publication ID	Publication ID	unique publication identifier
Submitted	Submitted	1 = true, 0 = false
Submitted Time	Submitted Time	timestamp
First Active	First Active	timestamp
Last Active	Last Active	timestamp
Publication Consent	Publication Consent	1 = true, 0 = false
Participatory Consent	Participatory Consent	1 = true, 0 = false
Approved By Moderator	Approved By Moderator	1 = true, 0 = false
Hidden By Moderator	Hidden By Moderator	1 = true, 0 = false
Identified	Identified	1 = true, 0 = false
Processed	Processed	1 = true, 0 = false
Index	Index	number
Geopoint	Geopoint	geographical coordinates
Zoom	Zoom	zoom level of the map
Latitude	Latitude	decimal degrees
Longitude	Longitude	decimal degrees
WKT	WKT	well-Known Text format (WGS 84)
GeoJSON	GeoJSON	GeoJSON formatted data

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PPGIS Metadata – Me an Nature in My Everyday Life

What_activities	What activities do you mostly do in this place?	true/false (summary)
Commuting_passing	Commuting / passing by	true/false
Enjoying_relaxing_recovering	Enjoying, relaxing, recovering mentally	true/false
Exercising_physical_activities	Exercising and physical activities (e.g. cycling, running)	true/false
Hiking_Camping	Hiking / Camping	true/false
Picking_ nature products	Picking mushrooms/berries/plants/other nature products	true/false
Pet_walking	Pet walking	true/false
Observing_nature_wildlife	Observing nature or wildlife	true/false
Meeting_friends_family	Meeting friends or family	true/false
Activities_with_children	Activities with children	true/false
Caring_for_nature	Caring for nature (e.g. picking up trash, invasive species prevention)	true/false
Gardening_Growing food	Gardening / Growing food	true/false
Water_activities	Water activities (e.g. swimming, boating)	true/false
Fishing_Hunting	Fishing / Hunting	true/false
Other	Other	true/false
Other_what	if other, what?	text.

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PPGIS Metadata – Me an Nature in My Everyday Life

How_often1	How often do you come here?	0 = Daily or almost daily, 1 = 2-3 times a week, 2 = Once a week, 3 = A couple of ti
How_often2	How often do you come here?	Daily or almost daily, 2-3 times a week, Once a week, A couple of times in a mont
Daily or almost daily	Daily or almost daily	true/false
2-3 times a week	2-3 times a week	true/false
Once a week	Once a week	true/false
A couple of times in a month	A couple of times in a month	true/false
Once a month	Once a month	true/false
Less frequently	Less frequently	true/false
Time_spent1	How much time do you usually spend in this place?	0 = Less than 0.5 hours, 1 = 0.5-1 hour, 2 = 1-2 hours, 3 = 2-3 hours, 4 = More thar
Time_spent2	How much time do you usually spend in this place?	less than 0.5 hours, 0.5-1 hour,1-2 hours, 2-3 hours, more than 3 hours
Less than 0.5 hours	Less than 0.5 hours	true/false
0.5 to 1 hour	0.5 to 1 hour	true/false
1 to 2 hours	1 to 2 hours	true/false
2 to 3 hours	2 to 3 hours	true/false
More than 3 hours	More than 3 hours	true/false
Modes_of_transportation	What modes of transportation do you most typically use to get here?	true/false (summary)
By_foot	By foot	true/false
By_bike	By bike	true/false
Public_transport	By public transportation	true/false
E-scooter	By e-scooter	true/false
Bycar	By car	true/false
Other_1	Other?	true/false

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Thank you!

Continue with practical ☺

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